Fiscal Year:	FY 2018	**Task Last Updated:**	FY 08/23/2017
PI Name:	Weitz, David Ph.D.		
Project Title:	Physics of Colloids in Space		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMPLEX FLUIDS/SOFT MATTER--Complex Fluids		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	weitz@seas.harvard.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	617 496-2842
Organization Name:	Harvard University		
PI Address 1:	Pierce 231		
PI Address 2:	29 Oxford Street		
PI Web Page:			
City:	Cambridge	**State:**	MA
Zip Code:	02138	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2009-10 Fluid Physics Rapid Turn Around Flight Opportunity
Start Date:	11/01/2013	**End Date:**	09/30/2019
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Meyer, William V	**Contact Phone:**	216-433-5011
Contact Email:	william.v.meyer@nasa.gov		
Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2019 per NSSC information (Ed., 10/12/18)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lu, Peter Ph.D. (Harvard University)		
Grant/Contract No.:	NNX13AQ48G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	NOTE: Continues "Physics of Colloids in Space," grant #NNX08AE09G. We will study the general collective behavior of colloidal particles, including the structures they form and the dynamics of the associated processes. As a result of longterm support from NASA, we have developed significant expertise in the formulation and characterization of systems of hard-sphere colloidal particles in a variety of solvents, and a host of techniques in the laboratory to characterize their structure and behavior, including several forms of microscopy, notably laser-scanning confocal microscopy, optical macro-photography, and laser-light scattering. Using chemistry again developed over the course of many years and supported by NASA, we can very carefully control the interaction potential between these colloidal particles. We can make the colloidal particles hard-spheres, interacting only insofar as to repel when contacting each other. Through the mechanism of depletion, we can make the particles attractive, and can exert very precise control over the range and strength of this attraction. In aqueous solvents, we can carefully control the screening-length, thereby controlling the range of repulsion of charged particles; and we can do the same with micelles in non-aqueous solvents. We have also developed new sample chamber technology, to improve the optics and mechanics for confocal microscopy, as well as control environmental factors and even exert electric fields. All of these ground-based studies have provided core technologies that we expect to apply to microgravity experiments, in particular the Light Microscopy Module/Advanced Colloids Experiment (LMM/ACE) set of experiments that involve microscopy aboard the International Space Station (ISS). The main goal of our work on colloids is to continue to explore fundamental science through the behavior of these well characterized systems, and apply the lessons directly to very similar systems in the lab with important industrial applications. Furthermore, we are using the same technology and techniques to explore materials beyond rigid spherical colloids that diffuse: in particular, self-propelled “colloids” in the form of swimming bacteria; flexible and deformable “colloids” in the form of micron-scale emulsion droplets, and the dynamics of charged systems in the presence of large external fields. This provides a way to apply directly the expertise we have developed to design and support ISS-based research, and has the potential to give significant new insight and direction for future experiments in microgravity. The work that we will do for the next 5 years primarily involves supporting our current set of experiments currently onboard the ISS, Binary Colloidal Alloy Test (BCAT) BCAT-3, BCAT-4, BCAT-5, and BCAT-6 (subsequently referred to collectively as BCAT), as well continuing to develop the hardware, software, and samples to launch as part of the set of upcoming experiments ACE-M2, ACE-M2R and ACE-confocal (subsequently referred to collectively as ACE). To do this properly, we must conduct a range of ground-based experiments, which allows the development of relevant technologies and expertise in order to successfully conduct flight experiments aboard the ISS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	These experiments are extremely important in determining the longer term shelflife stability of commercial products, and we plan to continue our work with some of our industrial contacts and colleagues, including Dr. Matthew Lynch at Procter and Gamble, to extend the utility of what we learn to real products. Our studies could also contribute significant understanding to the stability of many products that will likely be used for extended stays in space.
Task Progress:	We have made substantial progress toward a number of goals, including our first published paper that includes data from our flight experiments onboard the International Space Station [Kodger, Lu, Wiseman and Weitz, "Stable, Fluorescent Polymethylmethacrylate Particles for the Long-Term Observation of Slow Colloidal Dynamics," Langmuir (2017)], co-authored with an astronaut flight-crew member who collected our data. We have further submitted a second publication on improved image analysis methods which we intend to apply to data from upcoming microscopy experiments, specifically the Advanced Colloid Experiments (ACE) being run inside the LMM (Light Microscopy Module). We have provided significant guidance on the upcoming set of experiments that at the moment have been designated as ACE-M2R. The first set of experiments involve re-imaging some of the samples that we examined as part of ACE-M2, several years ago, without remixing, in the LMM microscope with bright-field and fluorescence microscopy modalities. This may give insight into the long-term evolution of the samples in the microgravity environment, which may shed light on the possibility of aging and slow phase separation in samples that were observed to have formed gels several years ago. If these experiments are successful, we may be able to make far more definitive statements on long-term stability of these systems, which is both quite important from the standpoint of fundamental science, and may have practical applications in the development of a number of household products favored by industry. Moreover, because this first stage of ACE-M2R will occur in existing facilities with samples already on orbit, it is a “zero-upmass” investigation that can test an important scientific question at extremely low marginal cost. We then intend to examine the same samples with the newly-launched confocal microscope facility being added to LMM, to gain information on 3D structure in our samples and evolution over time. We have complied fully and met or exceeded all of our requirements for our grant and augmentation in 2017, promptly responding and delivering technical solutions to all requests that we have received from both NASA and its contractors involved in our experiment. With our upcoming experiments with ACE-M2R and beyond, we believe that we are in position to capitalize on years of hard work invested into our colloid experiments, which themselves have already yielded the first of several publications. We are very excited by the prospect of the new data we expect to collect as part of the ACE-M2R / confocal experiments, for which we continue to make ongoing preparations and will support when run. We look forward to continuing our successful partnership with NASA in upcoming years, with colloid experiments on the ground and in microgravity, aboard the International Space Station.
Bibliography Type:	Description: (Last Updated: 06/13/2025)
Articles in Peer-reviewed Journals	Kodger TE, Lu PJ, Wiseman GR, Weitz DA. "Stable, fluorescent polymethylmethacrylate particles for the long-term observation of slow colloidal dynamics." Langmuir. 2017;33(25):6382–9. https://doi.org/10.1021/acs.langmuir.7b00852 , May-2017